
Software Requirements Specification

for Online Food Ordering System

Version 1.0 approved

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Revision History

Version	Date	Description	Author
1.0	2025-04-14	Initial Draft	Yashraj Upadhyay
1.1	2025-04-14	Incorporated Feedback	Harshvardhan Jadeja

1. Introduction

1.1 Purpose

This document defines the software requirements for the Online Food Ordering System developed using ASP.NET C#. It is intended for developers, testers, stakeholders, and end-users to understand the system scope, functionality, and architecture.

1.2 Document Conventions

- Requirements are labeled as REQ-x (e.g., REQ-1)
- High, Medium, Low priority tags used
- Bold is used for section titles, italics for emphasis

1.3 Intended Audience and Reading Suggestions

- **Developers:** For architecture and implementation reference
- **Testers:** For validating requirements
- **Project Managers:** For tracking progress

- **End-Users:** To understand system capabilities
Start with the introduction, followed by system features and interface requirements.

1.4 Project Scope

The system allows users to register/login, view categorized menus from restaurants, order food, and generate bills. Admins can manage restaurants, categories, items, and orders. The system boosts operational efficiency and improves user experience.

Additionally, the system allows users to book a table online at a preferred restaurant, date, and time.

1.5 References

- ASP.NET Web Forms Documentation
 - Microsoft SQL Server Docs
 - Bootstrap Documentation
 - IEEE SRS Template
-

2. Overall Description

2.1 Product Perspective

This web-based application is standalone but modular, with separate panels for users and admins. It includes database interaction, session management, and responsive design.

2.2 Product Features

- User registration/login
- Menu browsing and filtering by category
- Cart management and order placement
- Admin dashboard for content management

- Online table reservation with date/time selection and confirmation

2.3 User Classes and Characteristics

User Class	Description	Technical Expertise
Admin	Manages content, orders, and reservations	Moderate
Registered User	Orders food, books tables, manages cart	Basic
Guest	Views menu (limited access)	Basic

2.4 Operating Environment

- OS: Windows 10 or above
- Web Browser: Chrome, Firefox, Edge
- Server: IIS / Visual Studio Localhost
- Backend: ASP.NET with C#, .NET Framework
- DBMS: Microsoft SQL Server

2.5 Design and Implementation Constraints

- Use of ASP.NET Web Forms or MVC
- SQL Server integration
- Must follow .NET design principles
- Data encryption and secure login

2.6 User Documentation

- User manual (PDF)
- Admin guide
- Contact Us

2.7 Assumptions and Dependencies

- Users have internet access
 - System uses SQL Server 2019+
 - Application hosted on IIS or Azure
-

3. System Features

3.1 User Module

3.1.1 Description and Priority

Allows users to browse the menu, add to cart, place orders, and book tables. (Priority: High)

3.1.2 Stimulus/Response Sequences

User logs in → Views menu → Selects items → Places order → Receives confirmation

User logs in → Navigates to table booking → Chooses restaurant/date/time → Confirms booking

3.1.3 Functional Requirements

- **REQ-1:** The system shall allow user registration and login
- **REQ-2:** The user shall be able to browse food items by category
- **REQ-3:** The system shall allow users to add items to a cart

- **REQ-4:** The system shall allow users to book tables by selecting restaurant, date, time, and number of guests

3.2 Admin Module

3.2.1 Description and Priority

Admin can manage categories, items, orders, users, and reservations. (Priority: High)

3.2.2 Stimulus/Response Sequences

Admin logs in → Adds/Edits categories → Views order reports → Manages table bookings

3.2.3 Functional Requirements

- **REQ-5:** The admin shall log in with secure credentials
 - **REQ-6:** The admin shall add/edit/delete categories
 - **REQ-7:** The admin shall manage menu items
 - **REQ-8:** The admin shall view and update orders
 - **REQ-10:** The admin shall view and manage table booking requests
-

4. External Interface Requirements

4.1 User Interfaces

- Responsive design using Bootstrap
- Navigation menu for category filtering
- DataList/GridView for displaying items
- Form inputs for login/registration
- Table booking form with dropdowns for restaurant, date picker, time slots, and guest count

4.2 Hardware Interfaces

- Web client via PC/mobile browser
- Server hosted on Windows with IIS

4.3 Software Interfaces

- ASP.NET (Frontend + Logic Layer)
- SQL Server for DB operations
- .NET Framework

4.4 Communications Interfaces

- HTTP/HTTPS protocol
 - AJAX for asynchronous updates
 - SQL for DB queries
-

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- System should respond within 2 seconds for 90% of requests
- Capable of handling 100 concurrent users

5.2 Safety Requirements

- Regular backups of database
- Prevent duplicate transactions

5.3 Security Requirements

- Role-based access (Admin/User)
- Password encryption
- Session timeout for inactivity

5.4 Software Quality Attributes

- Usability: Mobile-friendly UI
 - Maintainability: Modular code structure
 - Reliability: 99.9% uptime target
 - Testability: Unit testing and user acceptance testing
-

6. Other Requirements

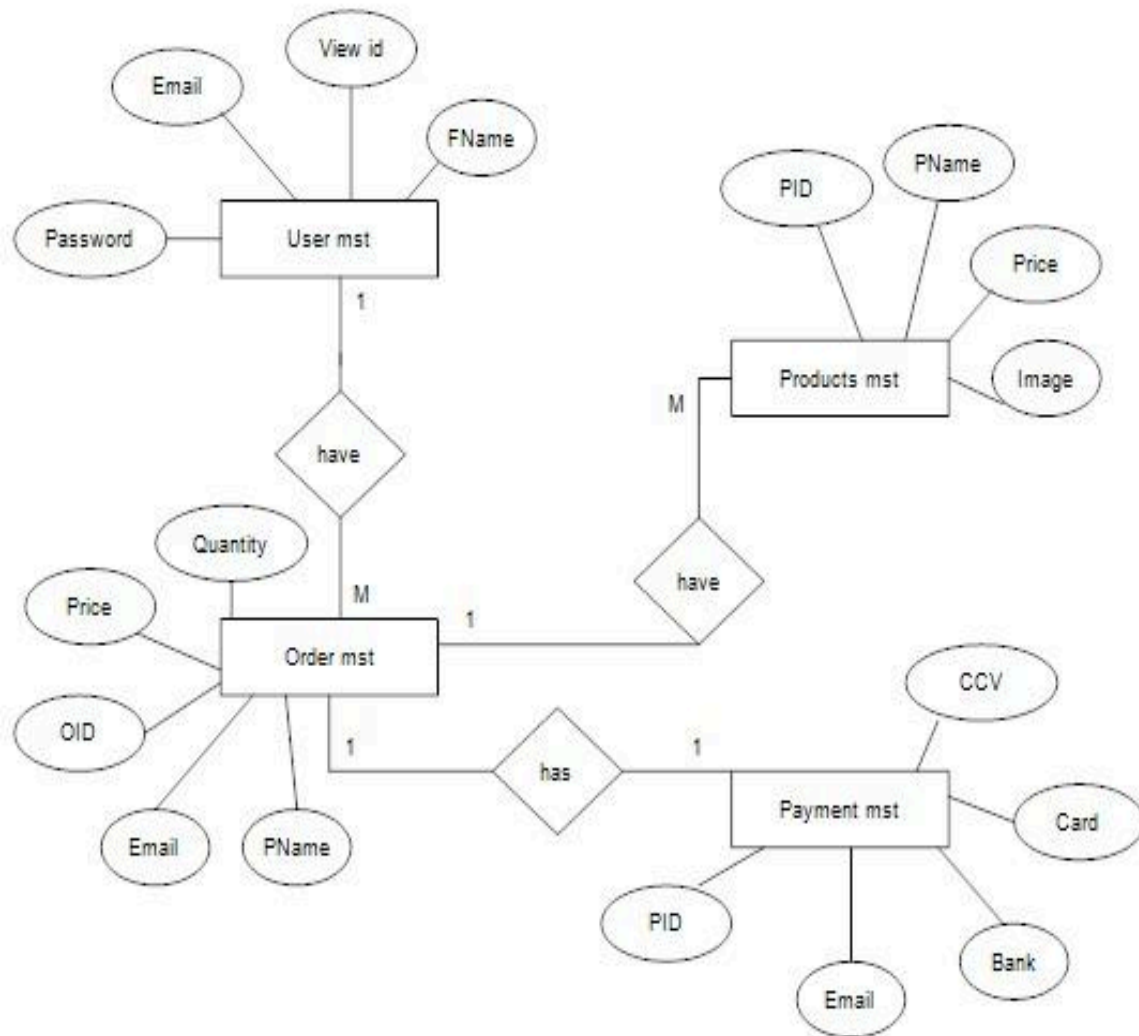
- Support for multiple restaurant menus in future versions
 - Multilingual support as a future enhancement
-

Appendix A: Glossary

- **ASP.NET:** A Microsoft framework for building dynamic web apps
 - **Cart:** Temporary selection of items for purchase
 - **SQL Server:** Relational database used for backend storage
 - **Booking:** Reserving a table at a restaurant online
-

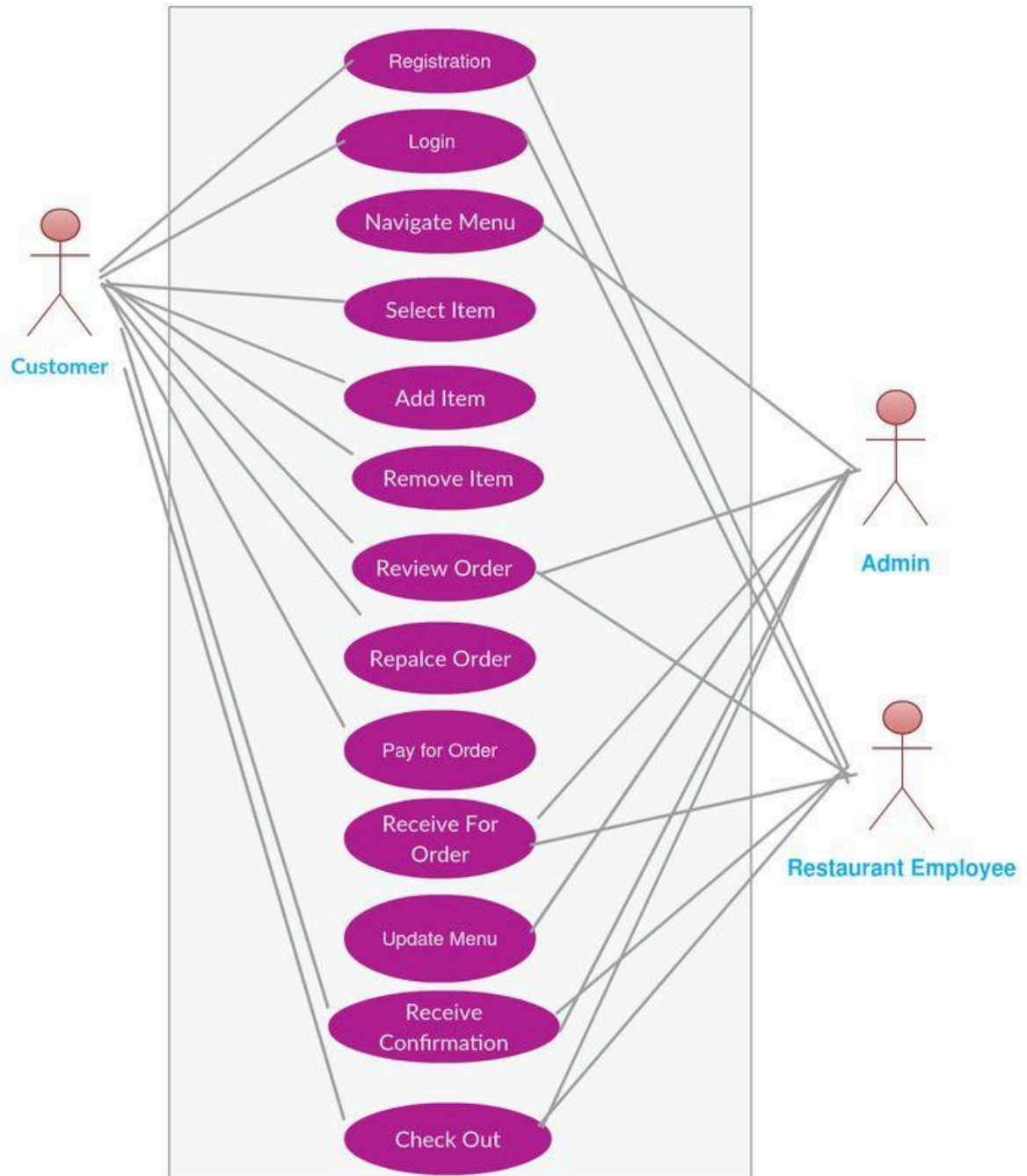
Appendix B: Analysis Models

1) ER Diagram



2) Use Case Diagram

Online Food Ordering System



3) Data Flow Diagram

I) Level Zero DFD

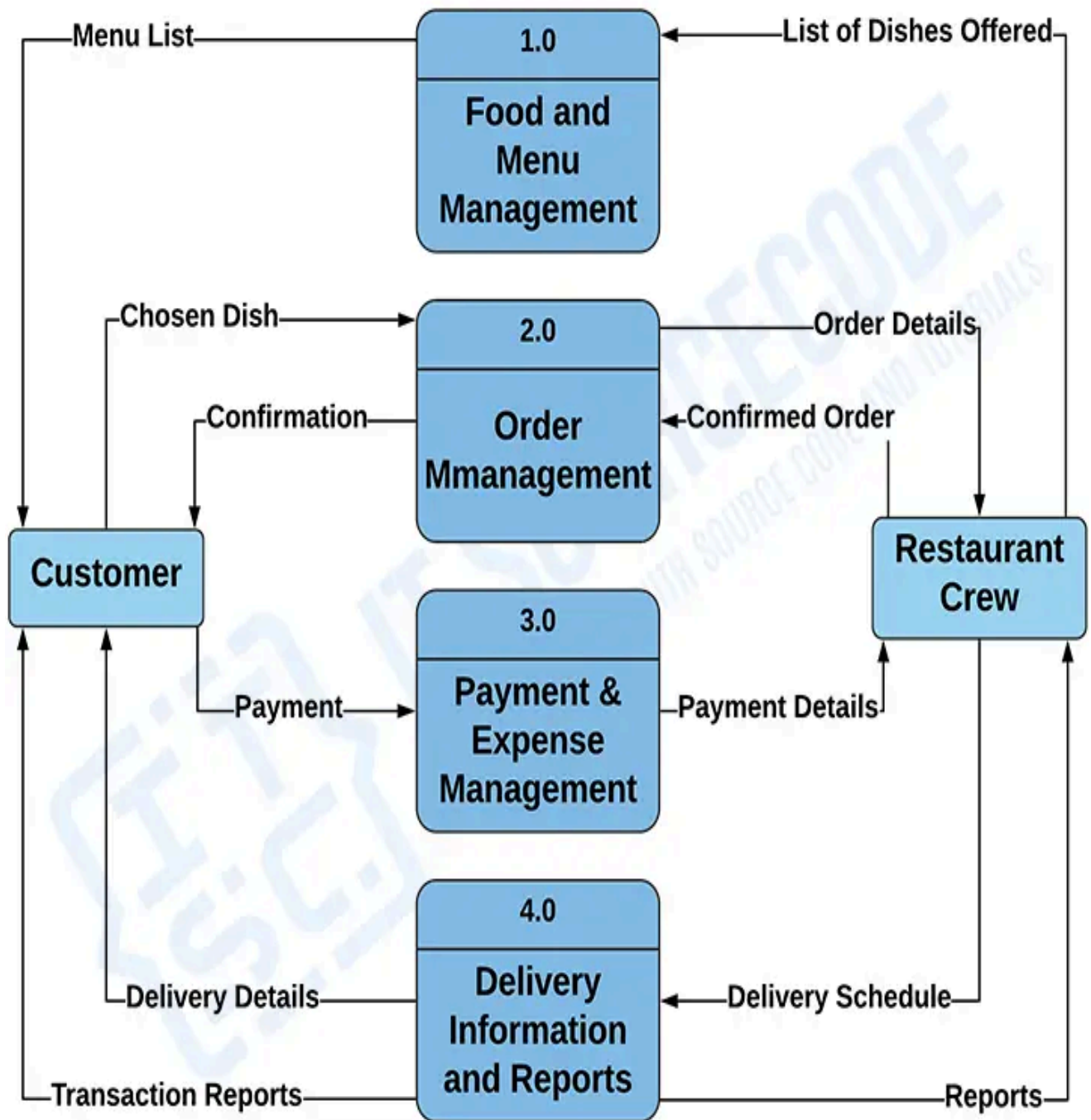
RESTAURANT MANAGEMENT SYSTEM



DATA FLOW DIAGRAM LEVEL 0

II) Level One DFD

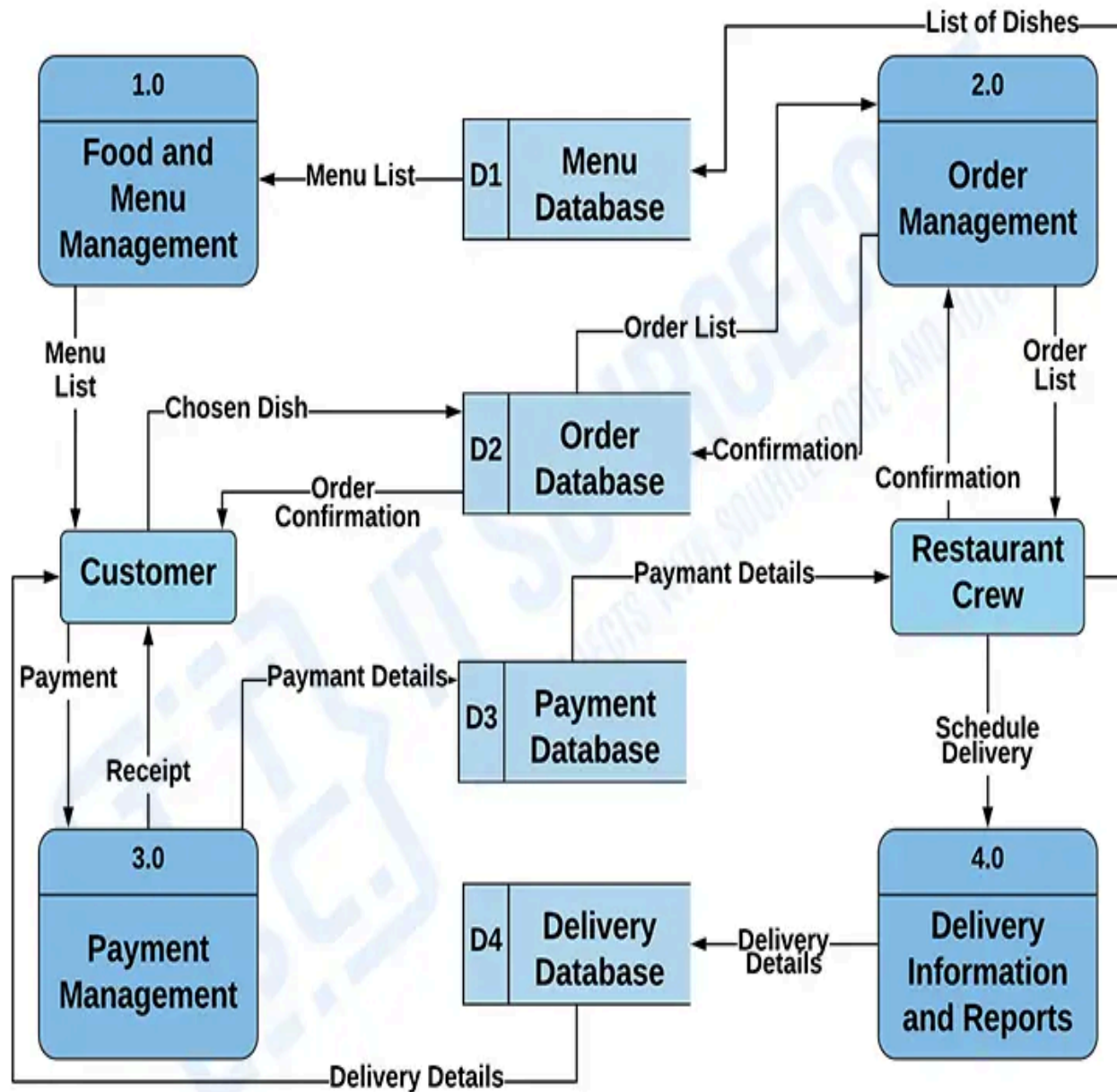
RESTAURANT MANAGEMENT SYSTEM



DATA FLOW DIAGRAM LEVEL 1

III) Level Two DFD

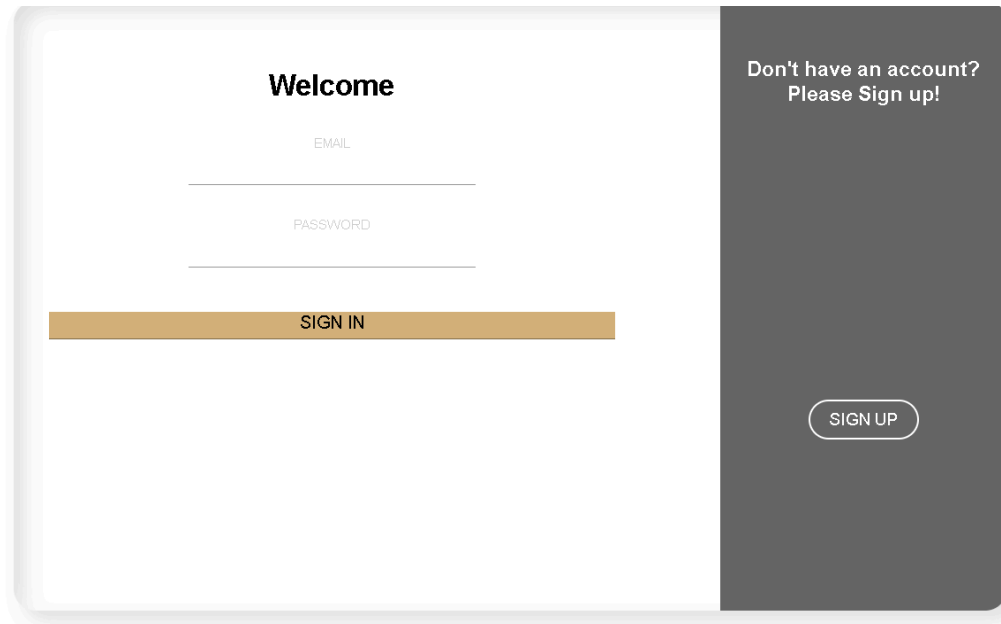
RESTAURANT MANAGEMENT SYSTEM



DATA FLOW DIAGRAM LEVEL 2

UI Design

1. Login page



The login page features a white main area and a dark gray sidebar. The white area contains a 'Welcome' heading, followed by input fields for 'EMAIL' and 'PASSWORD', and a prominent orange 'SIGN IN' button. The dark gray sidebar contains the text 'Don't have an account? Please Sign up!' and a white 'SIGN UP' button.

Welcome

EMAIL

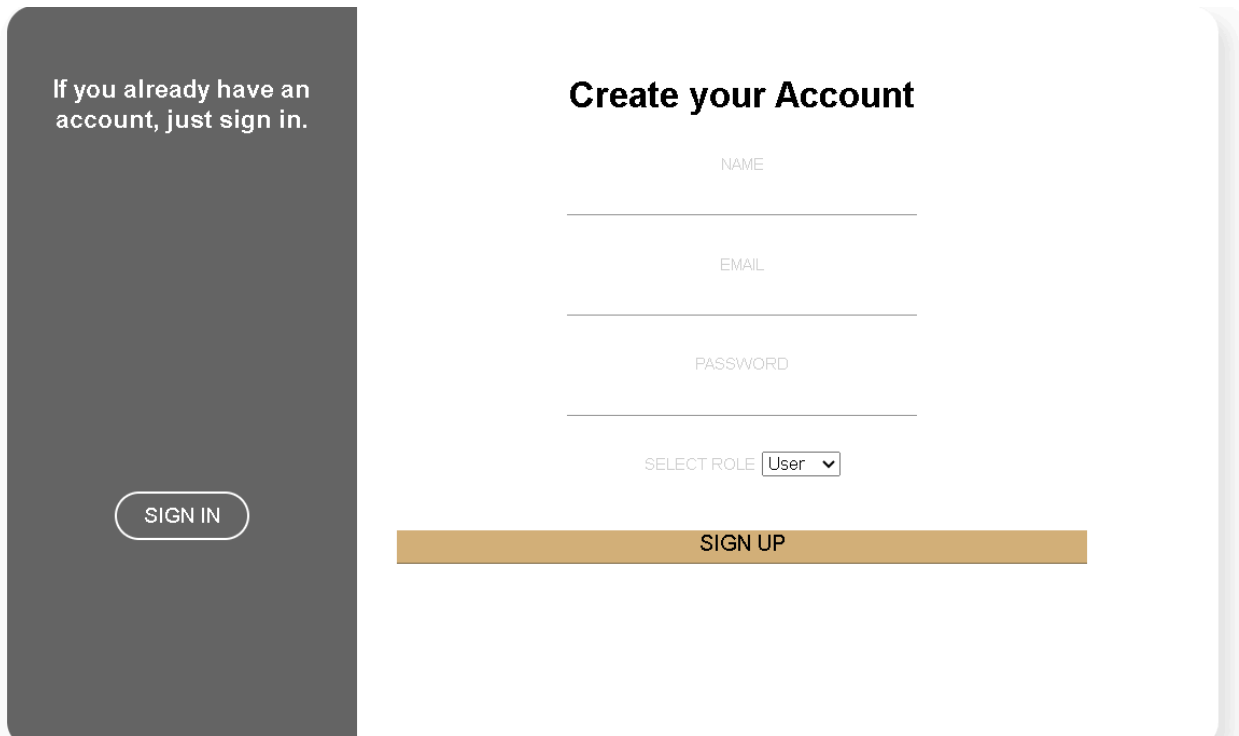
PASSWORD

SIGN IN

Don't have an account?
Please Sign up!

SIGN UP

2. Sign Up Page



The sign up page features a dark gray sidebar and a white main area. The sidebar contains the text 'If you already have an account, just sign in.' and a white 'SIGN IN' button. The white area contains a 'Create your Account' heading, followed by input fields for 'NAME', 'EMAIL', and 'PASSWORD', a 'SELECT ROLE' dropdown menu with 'User' selected, and a prominent orange 'SIGN UP' button.

If you already have an account, just sign in.

SIGN IN

Create your Account

NAME

EMAIL

PASSWORD

SELECT ROLE User ▼

SIGN UP

3. Index Page

**Restoran**

[HOME](#) [ABOUT](#) [SERVICE](#) [MENU](#) [PAGES](#) [CONTACT](#) [BOOK A TABLE](#)

Welcome, Guest!

Enjoy Our Delicious Meal

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[BOOK A TABLE](#)



**Master Chefs**

Dignidolor digne ipsum est. Aliq digne nam digne est. Clio erat ipsum est lorem est, sed erat lorem est digne ipsum nam digne erat digne

**Quality Food**

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**Online Order**

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**24/7 Service**

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About Us

Welcome to Restoran

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15

Years of Experience

50

Popular

[READ MORE](#)

Food Menu

Most Popular Items

 Popular Breakfast  Special Lunch  Family Dinner

**Chicken Burger**

digne digne of Tempor Lorem Digne Inc Digne

\$ 3.35

**Chicken Burger**

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**Chicken Burger**

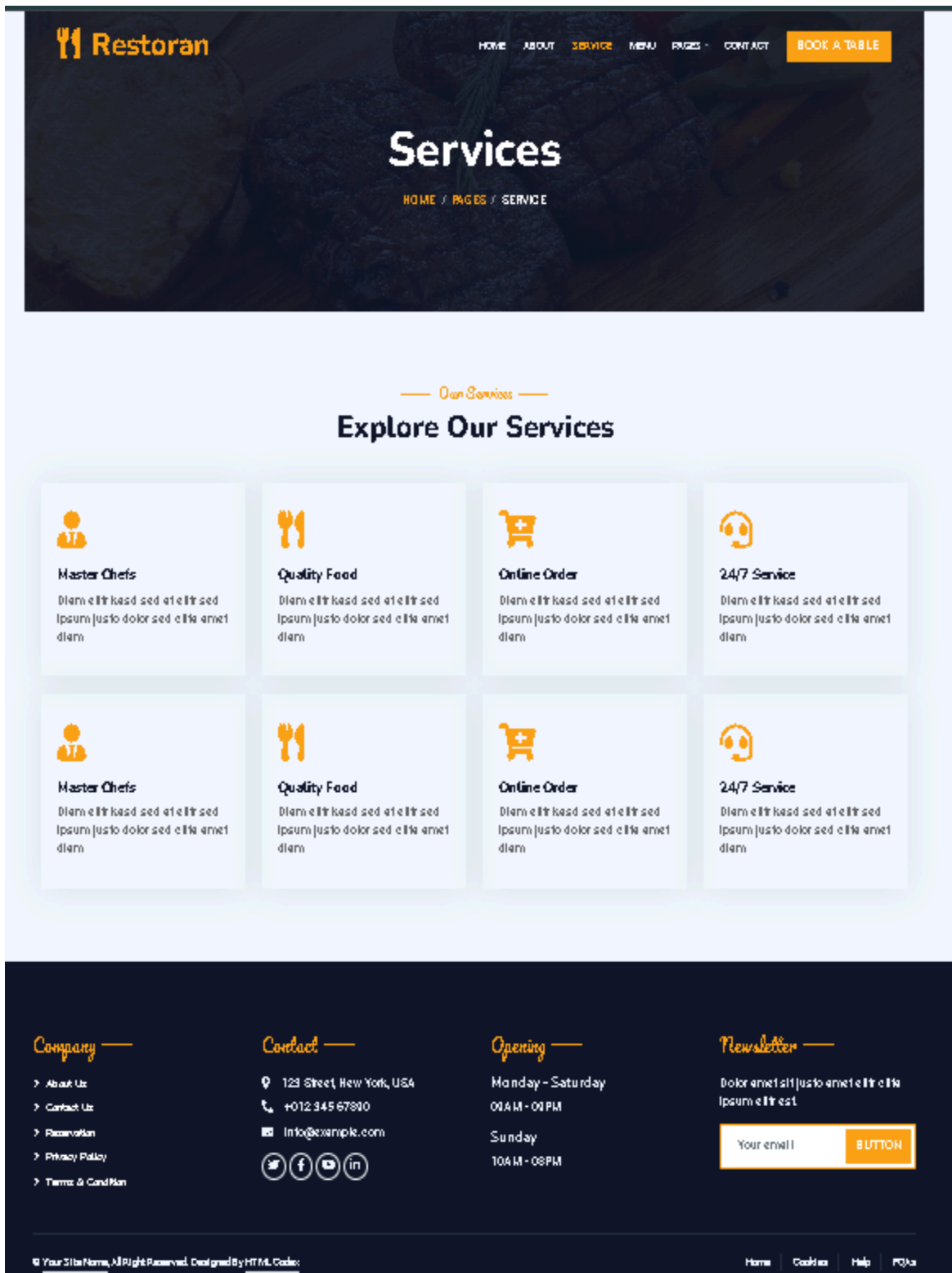
digne digne of Tempor Lorem Digne Inc Digne

\$ 3.35

2. About Us Page



3. Our Services



4. Product

 Restoran

HOMEABOUTSERVICEMENU*PAGES* *CONTACTBOOK A TABLE

Product

HOME / PAGES / PRODUCT

Breakfast ▾



Thepla

Price: ₹100

VIEW DETAILSADD TO CART



Dhokla

Price: ₹100

VIEW DETAILSADD TO CART



Aloo Paratha

Price: ₹120

VIEW DETAILSADD TO CART

Next

Company

- > About Us
- > Contact Us
- > Reservation
- > Privacy Policy
- > Terms & Condition

Contact

📍 123 Street, New York, USA

☎ +012 345 67890

✉ info@example.com



Opening

Monday - Saturday

09AM - 09PM

Sunday

10AM - 08PM

Newsletter

Dolor amet sit justo amet elitr clita ipsum elitr est.

SIGNUP

@ Your Site Name, All Right Reserved. Designed By [HTML Codex](#)

HomeCookiesHelpFAQs

5. Add To Cart

 Restoran

HOMEABOUTSERVICEMENU[▼]PAGES[▼]CONTACTBOOK A TABLE

Add To Cart

HOME / PAGES / PRODUCT

Your Cart Items



Pizza

Pizza is a popular Italian dish consisting of a baked flatbread topped with tomato sauce, cheese, and various toppings like vegetables, meats, and herbs. It has a crispy crust, gooey cheese, and a delicious blend of flavors. Available in different styles such as thin crust, deep-dish, and stuffed crust, pizza is loved worldwide. Versatile and customizable, it is a favorite comfort food enjoyed at gatherings, parties, and casual meals.

Qty: 1

Price: ₹499.00

Total: ₹499.00

Remove

Place Order

6. Address Details

 Restoran

HOMEABOUTSERVICEMENU[▼]PAGES[▼]CONTACTBOOK A TABLE

Address Detail

Address

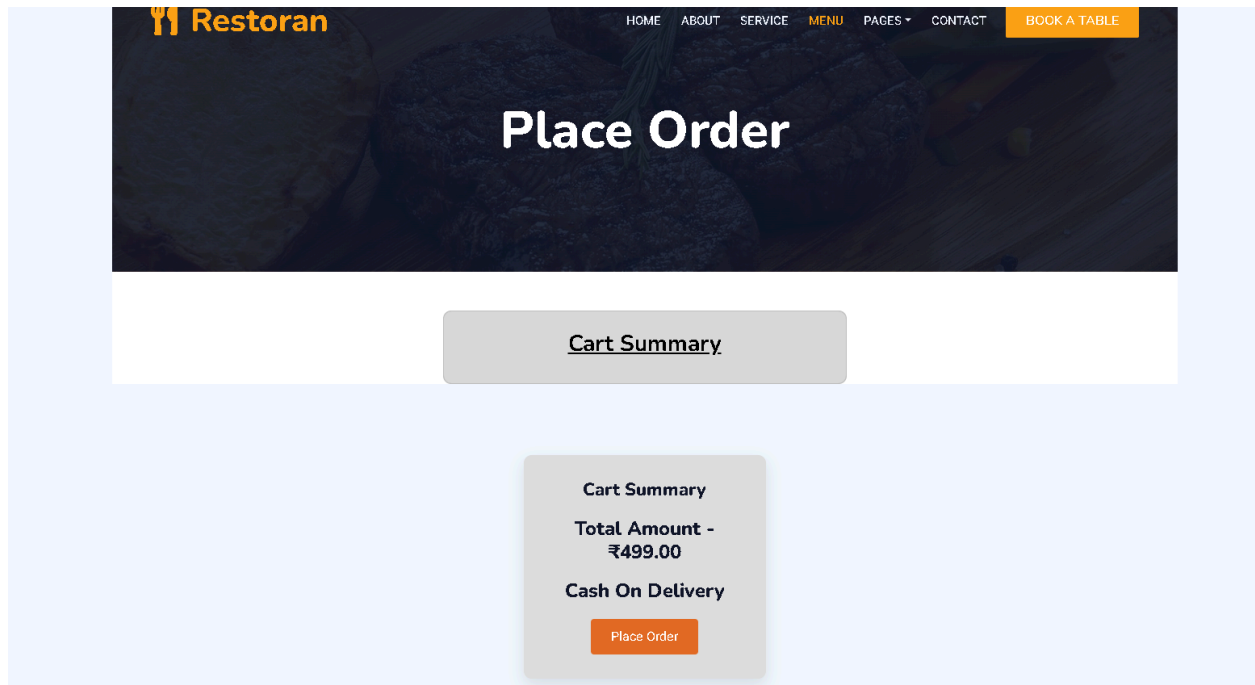
Your Name

Your Phone No.

Address

Place Order

7. Place Order



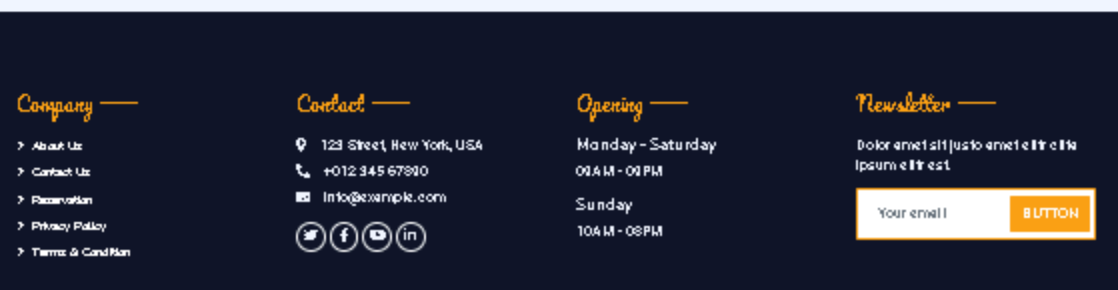
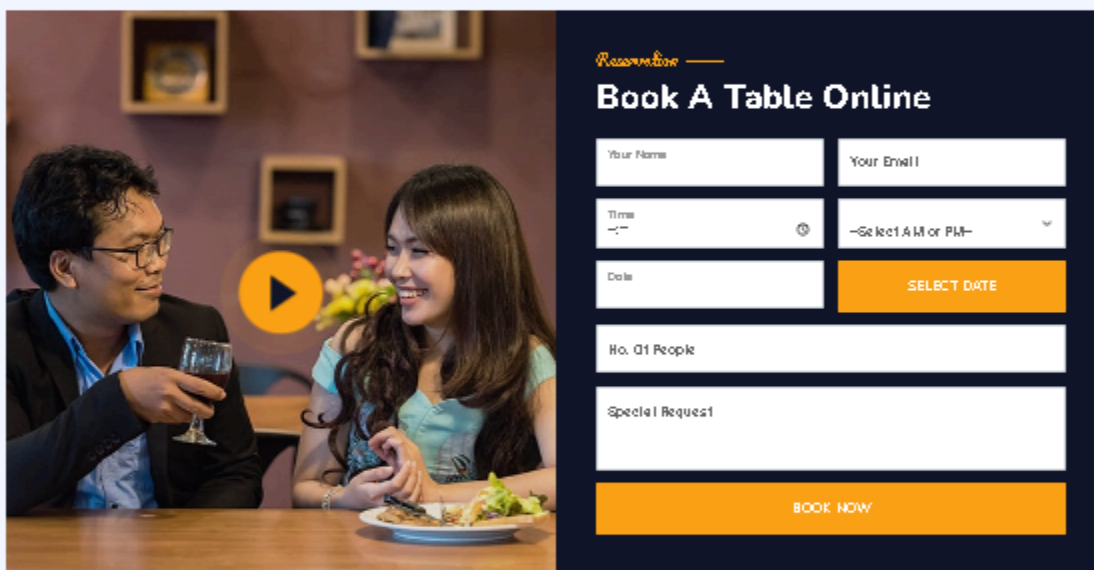
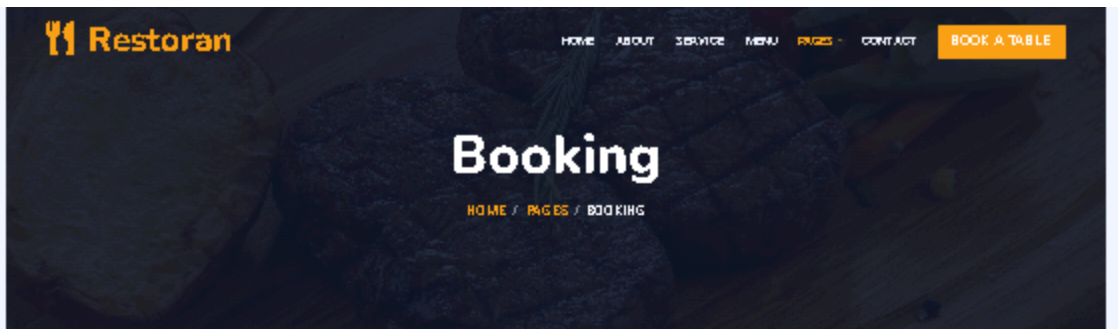
8. Order Confirmation

Order Placed Successfully!

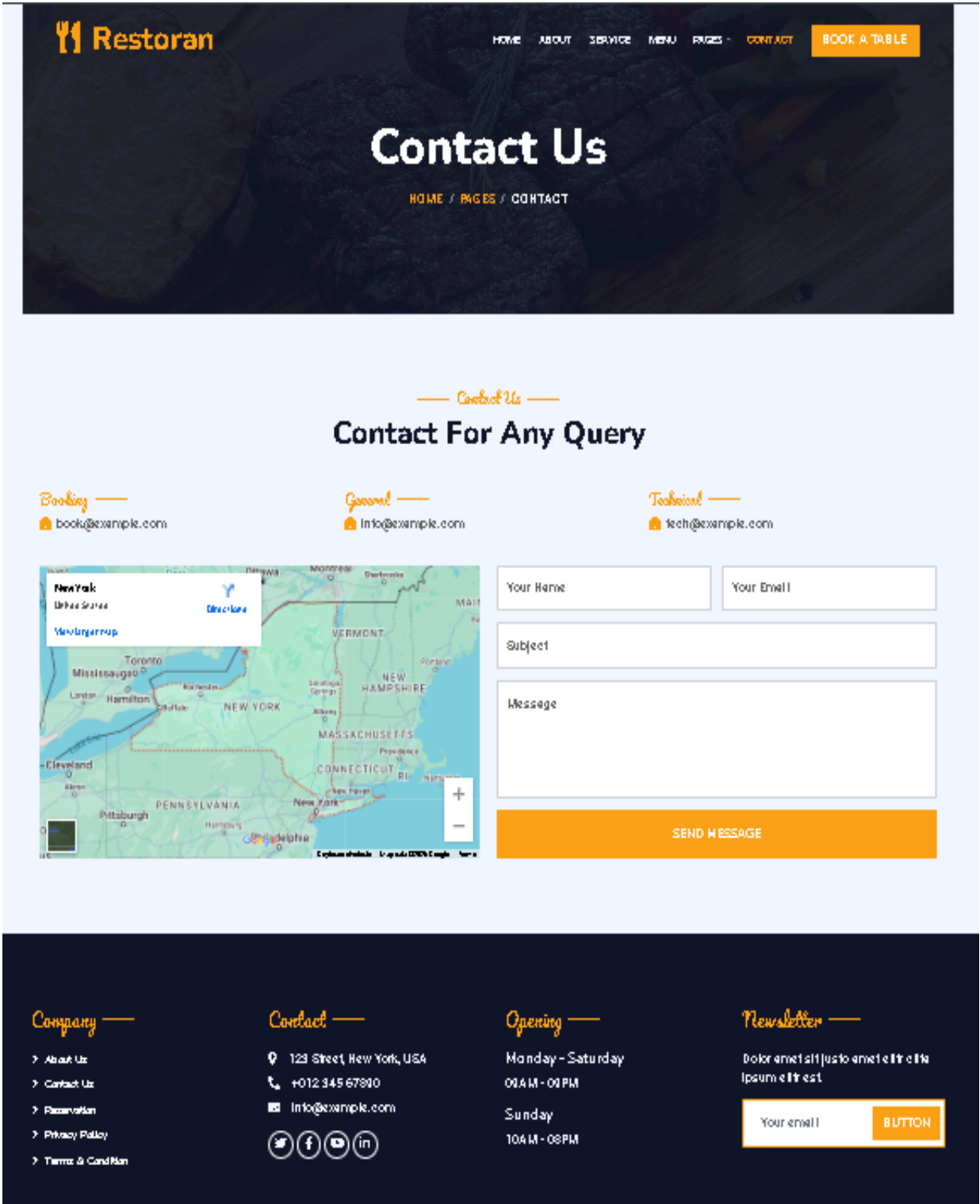
Thank you for your order. We'll deliver it soon!

[Continue Shopping](#)

9. Table Booking

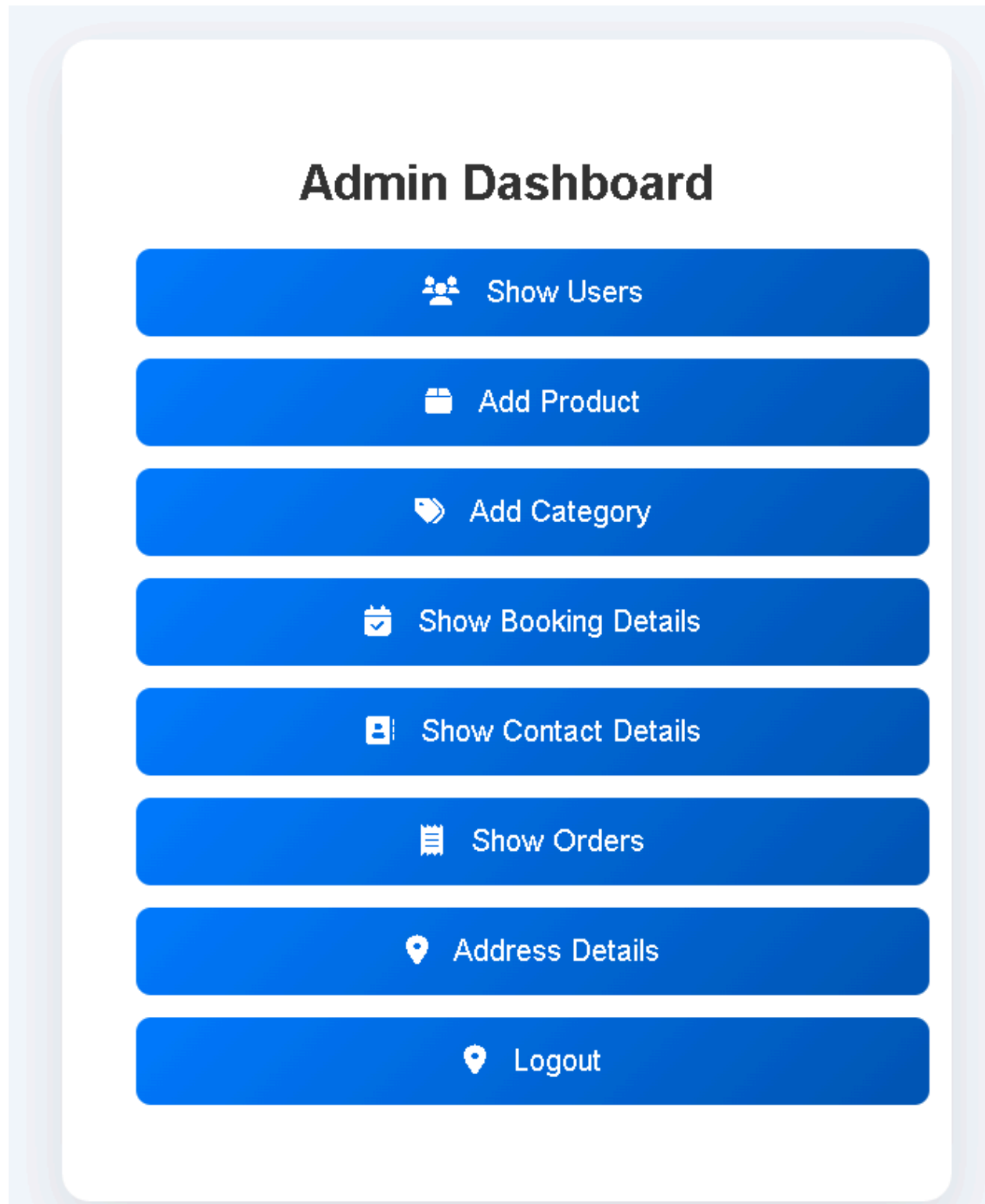


10. Contact Us



Admin Panel UI

1. Dashboard



2. Show Users

Manage Users

Id	Name	Email	Role	Password	Remove
1	Yashraj	yupadhyay404@rku.ac.in	User	123456	Remove
2	Harshvardhansinh	harshv@gmail.com	User	555666	Remove
3	Yashraj	upadhyayyashraj7@gmail.com	Admin	123456	Remove
6	siddh	siddh@gmail.com	User	123456	Remove
7	Disha	disha@gmail.com	User	123456	Remove
8	DD	d@gmail.com	User	123456	Remove
9	Deep	deep@gmail.com	User	123456	Remove
10	Sid	s@gmail.com	User	123456	Remove

[Report](#)

3. Add Product

Add Product

Select Category:

-- Select Category --

Product Name:

Price:

Upload Image:

[Choose File](#)

No file chosen

Description:

[Add Product](#)

[Report](#)

4. Add Category

Add Category

Category Name:

Add Category

5. Booking Table Details

Manage Bookings						
Booking ID	User Name	Email	Date	People	Special Request	Remove
1	yashraj	yupadhyay404@rku.ac.in	01-04-2025 22:17 AM	2	ednn	Remove
2	siddh	siddh@gmail.com	01-03-2025 03:03 AM	7	bhjbbbjh	Remove
3	Phone	phaone@gmail.com	18-03-2025 09:01 AM	8	jhh	Remove
4	hihihihi	p404@rku.ac.in	25-03-2025 12:14 AM	65	bhjbbbjh	Remove
5	Harshvardhansinh Jadeja	harubha521@gmail.com	23-02-2025 22:17 AM	7	No Mexican food and bill should be splitted equally	Remove
6	ijfios	hat@gmail.com	23-02-2025 22:17 AM	5	fwefw	Remove
8	yashraj	yupadhyay404@rku.ac.in	01-03-2025 03:03 AM	63	dadll	Remove
11	yashraj77	yupadhyay404@rku.ac.in	25-02-2025 01:10 PM	8	kjj	Remove

6. Contact Us Details

Manage Contact Details

ID	Name	Email	Subject	Message	Remove
1	yashraj	yupadhyay404@rku.ac.in	error	xyz	Remove
2	hasr	yup	jiwhf	abçajdjsifhwie	Remove

7. Show Orders

Manage Orders

Order ID	Cart ID	User ID	Product ID	Order Date	Status	Remove
57	40	7	43	15-04-2025	Pending	Remove
58	41	1	44	15-04-2025	Pending	Remove
59	42	1	42	15-04-2025	Pending	Remove
60	41	1	44	15-04-2025	Pending	Remove
61	42	1	42	15-04-2025	Pending	Remove
62	43	1	43	15-04-2025	Pending	Remove
65	45	8	43	15-04-2025	Pending	Remove
66	40	7	43	15-04-2025	Pending	Remove
67	44	7	42	15-04-2025	Pending	Remove
68	45	8	43	15-04-2025	Pending	Remove
69	46	8	44	15-04-2025	Pending	Remove
70	45	8	43	15-04-2025	Pending	Remove
71	46	8	44	15-04-2025	Pending	Remove
72	41	1	44	15-04-2025	Pending	Remove

8. Address Details

Address Details

ID	Name	Phone No	Address	Remove
4	yashraj upadhyay	09408122261	Gondal	Remove
8	yashraj upadhyay	0940812226	Bhavnath 2,Gondal	Remove
9	cafaf	7777777708	casas	Remove
10	fdada	4444444444	l,mcapko	Remove
11	aC	0078878990	CACC	Remove
12	yashraj upadhyay	7777777777	Bhavnath 2,Gondal	Remove
13	yashraj upadhyay	7894561238	Bhavnath 2,Gondal	Remove
14	jijijiji	7077894987	fewfaw	Remove
15	fasdfas	7410236589	njnn	Remove
16	yashraj upadhyay	7108898478	Bhavnath 2,Gondal	Remove
17	jijiji	7410236589	kookkoko	Remove
18	hdoahhuo	7894561238	kknkn	Remove
19	oijoiho	7777777777	,,kll	Remove
20	yashraj upadhyay	7894520316	Bhavnath 2,Gondal	Remove

Appendix C: Issues List

- TBD: Payment gateway integration
- TBD: Real-time order tracking
- TBD: Restaurant rating system
- Implement table availability logic and prevent double booking